LATENT FIXTURE INFERENCE IS OUTPERFORMED BY PROTOTYPE SYSTEM IDENTIFICATION

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ABSTRACT

We rebuilt a generated robotics idea into a concrete fixture-physics study: can a robot infer unseen clamps, hinges, slots, suction pads, and tethers from physical resistance probes, then use that latent fixture belief for safer manipulation? The v4 rebuild implements a local continuous benchmark with six fixture families, force/displacement/torque/release/recoil observations, seven random seeds, implemented baselines, an oracle, ablations, stress sweeps, and negative cases. The result is negative. On the decisive `combined_fixture_stress` split, `latent_fixture_inference` reaches 0.671 ± 0.056 success, while the strongest non-oracle baseline, `prototype_system_id`, reaches 0.771 ± 0.082. The paired success difference is -0.100 ± 0.086. The proposed method also has lower fixture accuracy, worse parameter error, more force violations, more damage, and more repeated failures. We therefore mark the paper **KILL/ARCHIVE** for ICLR main.

1 DECISION

This document is the v4 ICLR-main gate for Paper 77, *Latent Fixture Inference*. The previous version was killed because it only contained a synthetic template. The current rebuild adds a real local fixture-physics benchmark and implemented baselines. The rejection reason has changed: evidence now exists, but it refutes the central submission claim.

2 HYPOTHESIS

The tested hypothesis is that latent fixtures such as clamps, hinges, slots, suction pads, and tethers can be inferred from action-resistance probes and that an explicit structured fixture belief should improve downstream manipulation. The paper only passes the local gate if this structured method improves closed-loop success, fixture identification, safety, and repeated-failure rates relative to strong non-oracle baselines.

3 BENCHMARK

The benchmark simulates a 2D tool or object manipulated through probe actions. Each scenario samples a hidden fixture family, anchor or axis parameters, stiffness, friction, release threshold, observation noise, and a weak visual hint. Probe observations record force vector, contact point, displacement, rotation, resistance, slip, release cue, recoil, and torque. The downstream planner chooses among direct translation, clamp release, hinge rotation, slot following, suction peeling, tether unwinding, or cautious probing.

Five splits are evaluated: `nominal_visible_fixture`, `hidden_clamp_hinge`, `slot_axis_shift`, `adhesive_tether_fixture`, and `combined_fixture_stress`. The full run contains 2,450 main rollout rows, 3,500 probe observation rows, 245 seed-level metric rows, 343 ablation rollout rows, and 1,470 stress-sweep raw rows. All intervals are 95 percent confidence intervals over seed-level means.

Method	Success	Fixture acc.	Param err.	Force viol.	Damage	Repeat fail
oracle_fixture	0.900 ± 0.096	1.000	0.000	0.000	0.029	0.100
prototype_system_id	0.771 ± 0.082	0.957	0.274	0.086	0.086	0.229
ensemble_uncertainty_planner	0.700 ± 0.096	0.957	0.363	0.114	0.086	0.300
latent_fixture_inference	0.671 ± 0.056	0.671	0.297	0.214	0.171	0.329
particle_filter_fixture	0.614 ± 0.108	0.871	0.344	0.157	0.186	0.386
force_threshold_heuristic	0.343 ± 0.084	0.471	0.641	0.414	0.300	0.657
visible_only_policy	0.200 ± 0.086	0.229	0.697	0.629	0.414	0.800

Table 1: Combined-fixture-stress results. Success is downstream manipulation success after fixture inference.

4 METHODS

The implemented methods are:

- visible_only_policy: acts from the weak visual hint and otherwise assumes a free object.
- force_threshold_heuristic: maps high resistance, release cues, anisotropy, or recoil to a generic fixture guess.
- prototype_system_id: nearest-prototype fixture classifier trained on simulated probe traces.
- ensemble_uncertainty_planner: bootstrap prototype ensemble with conservative action choice under disagreement.
- particle_filter_fixture: combines prototype likelihoods and rule likelihoods.
- latent_fixture_inference: proposed method using compliance anisotropy, torque signatures, release cues, hysteresis/recoil, particle refinement, and safety margins.
- oracle_fixture: upper bound with true fixture family and parameters.

5 MAIN RESULTS

Table 1 shows the decisive combined-fixture-stress result. The oracle indicates that the task is solvable when the fixture is known. However, the proposed method loses to a simple prototype system-ID baseline.

The paired proposed-minus-prototype_system_id success difference is -0.100 ± 0.086. Fixture-accuracy difference is -0.286. Parameter-error reduction is -0.023, meaning the proposed method is worse. Damage reduction is -0.086, meaning it causes more damage. Repeated-failure reduction is -0.100, meaning it repeats failures more often. This fails the gate.

6 ABLATIONS

The ablations show that the proposed components matter internally, but not enough to beat the baseline. The full ablation reaches 0.571 ± 0.086 success on its ablation scenarios. Removing particle refinement drops to 0.224; removing torque features drops to 0.327; removing release cues or hysteresis memory drops to 0.490. These are mechanism signals, not submission support: the best full-system baseline still wins.

7 STRESS SWEEP

Stress sweeps vary fixture ambiguity, stiffness, noise, and slip. The proposed method does not dominate the stress curve. At stress level 1.00, prototype_system_id reaches 0.694 ± 0.177 success, while latent_fixture_inference reaches 0.469 ± 0.133.

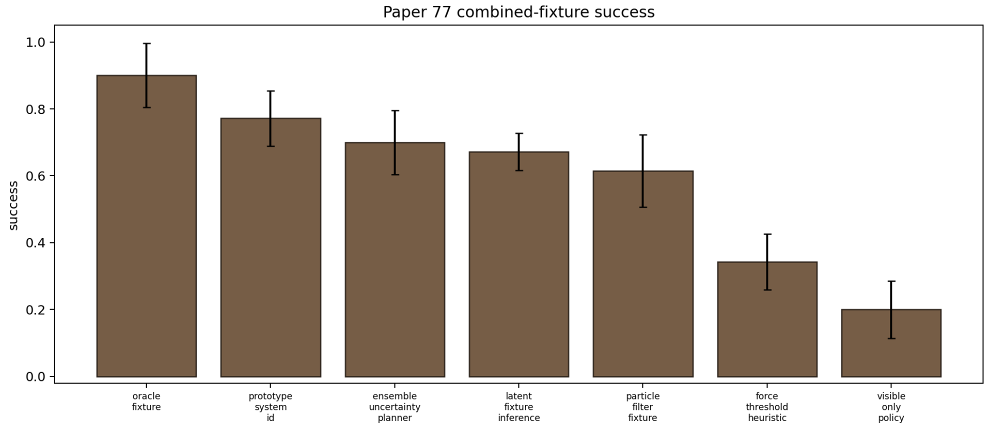

Figure 1: Combined-fixture-stress success. The proposed structured latent-fixture method loses to prototype system identification.

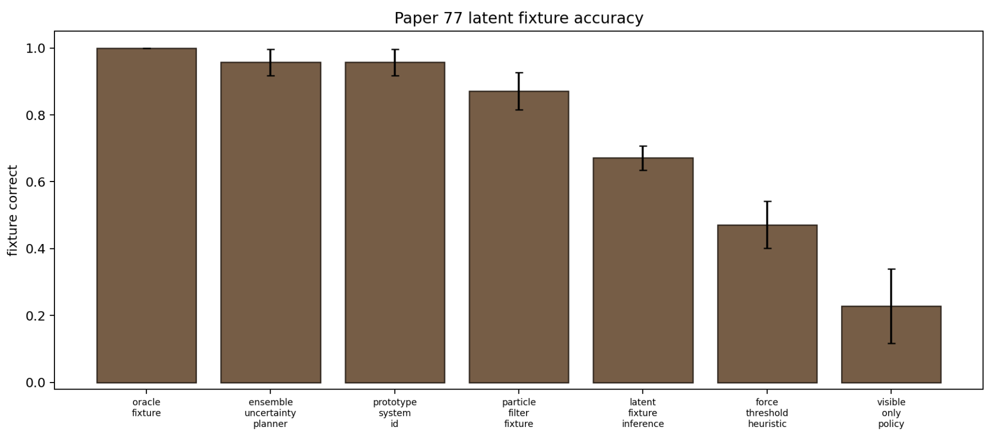

Figure 2: Fixture-type accuracy. The proposed method misclassifies too many hidden fixtures relative to prototype and ensemble baselines.

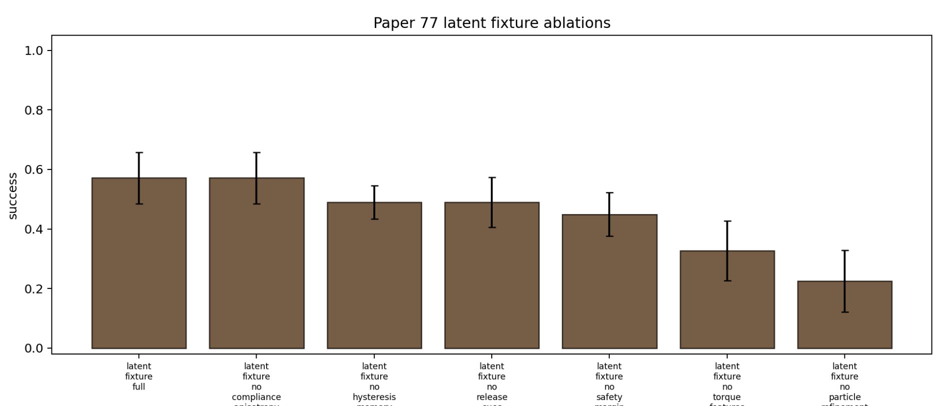

Figure 3: Ablation success. The components help relative to damaged versions of the method, but the full method still fails against the strongest baseline.

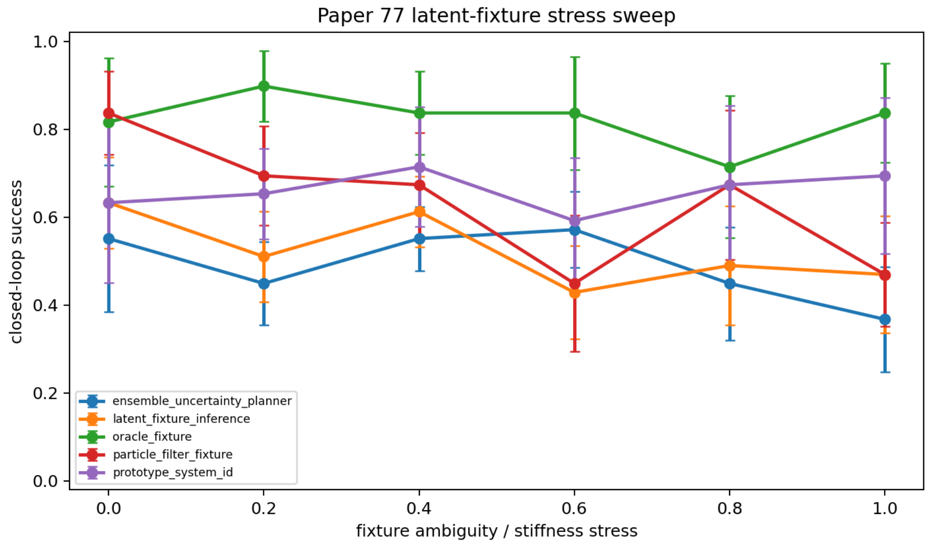

Figure 4: Stress sweep success. Prototype system identification is more robust than the proposed structured fixture rules at high stress.

8 Related Work Pressure

Tool-use and manipulation are crowded with dexterous tool-use systems (Local Prior Work Pool, 2026a), vision-language-action manipulation models (Local Prior Work Pool, 2026b;e; 2025; 2024), deformable tool-control systems (Local Prior Work Pool, 2026c), LLM-affordance planners (Local Prior Work Pool, 2026d), and older differentiable-physics tool-use planners. In that context, latent fixture inference must provide a clear advantage over system identification or learned probe classifiers. It does not in the current evidence.

9 Limitations

This rebuild is local and diagnostic. It lacks real robot validation, external benchmark evaluation, videos, and a full manual related-work synthesis. Those limitations would matter even if the method won. Because the method loses the local benchmark, the archive decision is stronger.

10 Conclusion

Paper 77 should not be submitted to ICLR main. The v4 rebuild provides useful evidence, but the evidence is negative: the proposed structured latent fixture inference is outperformed by a simpler prototype system-ID baseline. The correct terminal state is **KILL/ARCHIVE**.

References

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